

Lecture 16

The Families of Post-Quantum Cryptography

SEC 595: Encrypted Computing

Roadmap

This lecture is an unassessed survey and introduces no new mathematics. The next two lectures treat ML-KEM and ML-DSA in depth, and it would be easy to conclude that post-quantum cryptography *is* lattices. It is not. Section 1 explains why post-quantum cryptography is needed and Section 2 describes the NIST standardisation process. Sections 3–8 survey six families: lattice, hash, code, multivariate, isogeny and MPC-in-the-head. Section 9 compares them, and Section 10 examines two schemes that were broken in 2022 — by mathematics, not by quantum computers.

1. Why Post-Quantum At All

1.1 What Shor’s Algorithm Breaks

A large fault-tolerant quantum computer running Shor’s algorithm solves, in polynomial time:

- (i) integer factorisation;
- (ii) the discrete logarithm problem in $\mathbb{Z}_p^\times$;
- (iii) the discrete logarithm problem on elliptic curves.

This breaks RSA, Diffie–Hellman and ECDSA. It also breaks the classical homomorphic schemes of this course: RSA (factoring), ElGamal (DLog in $\mathbb{Z}_p^\times$) and Paillier (factoring $n = pq$) from Lecture 8, and BGN from Lecture 9, whose composite-order pairing group rests on factoring $n = q_1q_2$. The lattice family is the reason homomorphic encryption has a future.

1.2 What Grover’s Algorithm Does

Grover’s algorithm searches an unstructured space of size N in $O(\sqrt{N})$ steps. Against a λ -bit key it halves the exponent: 2^{128} work becomes about 2^{64} . The fix is to double the key length. AES-256 and SHA-384 are considered post-quantum-adequate; AES-128 is not.

Asymmetric cryptography is destroyed; symmetric cryptography is merely inconvenienced. So every family below builds public-key primitives from problems that look nothing like factoring.

1.3 Harvest Now, Decrypt Later

Confidentiality is retroactive: ciphertext recorded today can be decrypted whenever a quantum computer arrives. Anything that must stay secret for twenty years needs post-quantum protection now. Authentication is not retroactive: a signature verified today cannot be forged later. This

is why key establishment migrated first, and why ML-KEM was deployed in browsers well before signature migration began.

2. The Standardisation Process

2.1 The NIST Standards

NIST issued its call in 2016 and announced the first selections in July 2022. The state of play as of September 2026:

Standard	Scheme	Family	Status
FIPS 203	ML-KEM (Kyber)	lattice	final, 13 Aug 2024
FIPS 204	ML-DSA (Dilithium)	lattice	final, 13 Aug 2024
FIPS 205	SLH-DSA (SPHINCS ⁺)	hash	final, 13 Aug 2024
FIPS 206	FN-DSA (Falcon)	lattice	draft
—	HQC	code	selected 11 Mar 2025

Three of the five schemes — ML-KEM, ML-DSA and the draft FN-DSA — are lattice schemes: the family won. NIST nevertheless standardised SLH-DSA and selected the code-based HQC, *because* the family won. Relying on one family is a single point of failure.

2.2 The Additional Signatures Round

NIST then opened a separate call for signatures *not* based on structured lattices. Fourteen submissions reached round 2, and on 14 May 2026 nine advanced to round 3:

Scheme	Family
UOV, MAYO, QR-UOV, SNOVA	multivariate
MQOM, SDitH, FAEST	MPC-in-the-head
SQIsign	isogeny
HAWK	lattice, but a different lattice problem

HAWK is the only lattice entry. It rests on the lattice *isomorphism* problem rather than on LWE, and so fails differently.

3. Lattice-Based Cryptography

3.1 The Hard Problems

This is the family the course has built. Lecture 5 gave γ -SVP, GapSVP, BDD, SIS and q -ary lattices; Lecture 6 gave LWE, its ring and module variants, and MSIS.

Scheme	Rests on
ML-KEM	Module-LWE over $R_q = \mathbb{Z}_q[X]/(X^{256} + 1)$, $q = 3329$
ML-DSA	Module-LWE and Module-SIS, $q = 8380417$
FN-DSA	NTRU lattices, with Gaussian sampling over a trapdoor
HAWK	the module lattice isomorphism problem

ML-KEM and ML-DSA are the subjects of Lectures 17 and 18.

3.2 Why It Won

Balance. Lattice schemes are the only family with *simultaneously* small keys, small ciphertexts or signatures, and fast arithmetic — the last because of the NTT over R_q (Lecture 3). Every other family is lopsided: enormous keys, or enormous signatures, or slow by orders of magnitude.

3.3 What Is Risked

Structure. The security reductions are for *unstructured* lattices; ML-KEM uses module lattices over a cyclotomic ring, a tiny and highly symmetric subset. No known attack exploits that structure at these parameters. Attacks on *ideal*-lattice problems do exist; they do not extend to module lattices only because module lattices are less structured.

4. Hash-Based Signatures

4.1 No New Hard Problem

A hash-based signature is secure if the hash function is preimage- and collision-resistant. Nothing else is assumed: no number theory, no lattices, no problem invented for the purpose. This is the most conservative assumption in cryptography.

4.2 The Construction

Definition 1 (Lamport one-time signature). Choose secret random values x_0, x_1 and publish $H(x_0)$ and $H(x_1)$. To sign the bit b , reveal x_b .

Signing twice with the same key reveals both preimages — hence *one-time*. WOTS⁺ compresses this using hash chains.

A one-time key signs one message, so build 2^h of them and authenticate their public keys with a binary **Merkle tree**. The root is the long-term public key; a signature carries the one-time signature plus the h sibling hashes on the path to the root. (This is the same object used as a commitment in Lecture 2.)

Merkle schemes such as XMSS and LMS must remember which one-time keys have been used, and a restored backup can reuse one. SPHINCS⁺, standardised as SLH-DSA, removes the state by choosing the leaf pseudorandomly from a tree large enough that collisions are negligible. Statelessness is paid for in signature size.

4.3 The Trade

Scheme	Signature (bytes)
SLH-DSA “small”	7,856
SLH-DSA “fast”	17,088
ML-DSA	2,420
ECDSA	64

SLH-DSA public keys are **32 bytes**, the smallest in this lecture. Signing is one to two orders of magnitude slower than ML-DSA. This is not a choice for TLS. It is the right choice for firmware and root certificates: signed rarely, verified for decades, where being wrong is unrecoverable.

5. Code-Based Cryptography

5.1 Decoding a Random Linear Code

Definition 2 (Syndrome decoding). Given a parity-check matrix $H \in \mathbb{F}_2^{r \times n}$ and a syndrome s , find a low-weight e with $He^T = s$.

The problem is NP-hard in general. The best known algorithms, information-set decoding, are exponential, with the exponent essentially unchanged since Prange (1962). The trapdoor is a code the owner can decode (a binary Goppa code) disguised as a random one. Lecture 2 used Reed–Solomon codes for low-degree testing; here the sign is flipped, and an easy code is hidden inside a hard-looking one.

5.2 The Oldest Family

McEliece published this in 1978, one year after RSA. It has never been broken.

Scheme	Public key	Ciphertext
Classic McEliece (level 1)	261,120 B	96 B
HQC (level 1)	≈ 2 KB	≈ 4 KB
ML-KEM-512 (for scale)	800 B	768 B

A quarter-megabyte public key cannot go in a TLS handshake, but a 96-byte ciphertext suits a long-lived static key exchanged once out of band. NIST chose HQC, based on random quasi-cyclic codes, for its practical key size; it pays with a larger ciphertext and a younger security history.

6. Multivariate Cryptography

6.1 Oil and Vinegar

Definition 3 (The $\mathcal{MQ}$ problem). Given m quadratic polynomials in n variables over $\mathbb{F}_q$, find a common root.

Solving a random quadratic system is NP-hard, and Gröbner-basis methods are exponential in general. The trapdoor is *oil and vinegar*: split the variables into two sets and build the map so that no two oil variables ever multiply. Fixing the vinegar variables at random makes the system *linear* in the oil variables, which is easy to solve. Secret invertible linear maps disguise the split.

6.2 The Shape of the Trade

UOV signs in about **96 bytes**, comparable to ECDSA — the smallest signatures in this lecture. Public keys are the largest, hundreds of kilobytes. MAYO and SNOVA try to shrink the key while keeping the signature small. Four of the nine round-3 signature candidates are multivariate.

7. Isogeny-Based Cryptography

An **isogeny** is a structure-preserving map between elliptic curves. In the supersingular isogeny graph, the vertices are curves and the edges are isogenies of small degree. The graph is an expander, so random walks mix fast, and recovering a path between two given curves appears hard.

SQIsign, in round 3, has signatures of about **148 bytes** and is one to two orders of magnitude slower than ML-DSA at signing. It is the family’s survivor after the break of SIDH/SIKE (Section 10).

8. MPC-in-the-Head

Take a symmetric primitive — AES, a syndrome-decoding instance, or an $\mathcal{MQ}$ instance — and prove in zero knowledge that you know the secret key:

1. simulate a multiparty computation “in your head”;
2. commit to the views of the virtual parties, and open a subset;
3. remove the interaction with Fiat–Shamir.

The transcript is the signature. Zero-knowledge proofs are the subject of Lecture 22.

The assumptions are as conservative as for hash-based signatures, and keys are tiny: the public key is an AES plaintext/ciphertext pair or a syndrome. Signatures run to several kilobytes and signing is slow. FAEST, SDitH and MQOM are in round 3.

9. Comparing the Families

9.1 Sizes

Figures are for NIST security level 1, in bytes. Values marked $\approx$ are order-of-magnitude.

Family	Representative	Type	Public key	Ct / sig
Lattice	ML-KEM-512	KEM	800	768
Code	Classic McEliece	KEM	261,120	96
Code	HQC	KEM	$\approx 2,000$	$\approx 4,400$
Lattice	ML-DSA-44	sig	1,312	2,420
Lattice	FN-DSA-512	sig	897	666
Lattice (LIP)	HAWK	sig	$\approx 1,000$	555
Hash	SLH-DSA-128s	sig	32	7,856
Hash	SLH-DSA-128f	sig	32	17,088
Multivariate	UOV	sig	$\approx 10^5$	96
Isogeny	SQIsign	sig	≈ 65	148
MPC-in-head	FAEST	sig	≈ 32	several KB
Classical	ECDSA P-256	sig	64	64
Classical	RSA-2048	sig	256	256

Read the table by columns. No scheme is small in both columns and also fast; every family loses in a different corner. ML-DSA is the best at nothing and terrible at nothing, which is what a default has to be.

9.2 Maturity

Family	Attacked since	Record
Code-based	1978	no break of the original proposal
Multivariate	1988	many breaks, including Rainbow in 2022
Hash-based	1979	reduces to hash security; no break
Lattice	1996	no break at standardised parameters
Isogeny	2006	SIDH/SIKE broken in 2022
MPC-in-head	2007	reduces to symmetric primitives; no break

Time in the arena is evidence, not proof.

10. The Two 2022 Breaks

10.1 Rainbow

Rainbow, a layered oil-and-vinegar signature, was a NIST round-3 *finalist*. In 2022 Beullens published an attack recovering the level-1 secret key in about a weekend on a laptop, and Rainbow was eliminated. No quantum computer was involved: a structural weakness in the trapdoor had survived five years of public analysis. The scheme had always been insecure; we found out late.

10.2 SIKE

SIDH and its KEM SIKE had by far the smallest keys of any post-quantum scheme. In July 2022, weeks after SIKE advanced to round 4, Castryck and Decru published a key-recovery attack running in about an hour on one core. Attacks by Maino–Martindale and by Robert followed within a month. The attack used the auxiliary *torsion points* that SIDH publishes, via a 1997 theorem of Kani. The isogeny problem itself was not broken. SQIsign publishes no torsion points.

10.3 What the Breaks Teach

In both cases the hard problem stood and the **trapdoor** — the extra structure that makes the scheme usable — leaked.

1. **Diversity is insurance.** Had NIST standardised only lattices in 2022, there would be no fallback.
2. **Time in the arena is a security parameter.** Code-based cryptography is forty-eight years old; lattices cannot yet say that.
3. **Deploy hybrids.** TLS deployments combine X25519 with ML-KEM-768, so the session is secure if *either* component holds, for a few hundred bytes.

Summary

1. Shor breaks factoring and both discrete-log problems: RSA, ECDSA, and this course’s El-Gamal and Paillier (L8) and BGN (L9). Grover only halves symmetric security.
2. Of the four FIPS standards — three final, FN-DSA still a draft — three are lattice schemes. SLH-DSA and the selected HQC are insurance.
3. The families trade key size, signature size, speed and assumption age. None wins on all four; lattices lose least.

4. Hash-based and MPC-in-the-head schemes assume only symmetric primitives. Code-based is the oldest family; multivariate has the smallest signatures and the worst record.
5. Rainbow and SIKE fell to classical mathematics in 2022 because the trapdoor leaked. So standardise several families and deploy hybrids.

Discussion Questions

1. SLH-DSA signatures are three times the size of ML-DSA's and signing is far slower. NIST standardised it anyway, on the same day. Give the strongest argument for that decision, and the strongest against.
2. Classic McEliece has a 261 KB public key and a 96-byte ciphertext. Describe a deployment where that is the right trade, and one where it is absurd. What property decides it?
3. Both 2022 breaks left the hard problem intact and broke the trapdoor. Does that suggest where to look for the next break? Does the lattice family have an analogous exposed surface?
4. Hybrid X25519 + ML-KEM-768 costs a few hundred bytes per handshake. When should a hybrid be retired, and who decides?
5. Harvest-now-decrypt-later argues for migrating encryption before signatures. Identify a setting where that priority is backwards.